

Eugene Moshchyn

Dr. Qi Cheng

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Assignment 5

Exercise 9.3-9 Describe an $O(n)$ -time algorithm that, given a set S of n distinct numbers and a positive integer $k \leq n$, determines the k numbers in S that are closest to the median of S .

Solution

- Use SELECT to find the median m of S .
- For each element $x \in S$, compute its distance from the median:

$$d(x) = |x - m|.$$

- Use SELECT again on the set of distances $\{d(x) : x \in S\}$ to find the k th smallest distance, say δ .
- Output all elements whose distance from the median is $\leq \delta$ to make a total of k elements.

Each step takes linear time, so the overall running time is $O(n)$.

Exercise 9.3-10 Let $X[1 : n]$ and $Y[1 : n]$ be two arrays, each containing n numbers already in sorted order. Give an $O(\lg n)$ -time algorithm to find the median of all $2n$ elements in arrays X and Y . Assume that all $2n$ numbers are distinct.

Solution

```

TWOARRAYMEDIAN( $X, Y, n$ )
1   $median = \text{FINDMEDIAN}(X, Y, n, 1, n)$ 
2
3  if  $median == \text{NotFound}$ 
4       $median = \text{FINDMEDIAN}(Y, X, n, 1, n)$ 
5
6  return  $median$ 

```

```
FINDMEDIAN( $X, Y, n, low, high$ )
1  if  $low > high$ 
2      return NotFound
3
4   $k = \lfloor (low + high)/2 \rfloor$ 
5
6  if  $k == n$  and  $A[n] \leq B[1]$ 
7      return  $A[n]$ 
8
9  if  $k < n$  and  $B[n - k] \leq A[k]$  and  $A[k] \leq B[n - k + 1]$ 
10     return  $A[k]$ 
11
12 if  $A[k] > B[n - k + 1]$ 
13     return FINDMEDIAN( $A, B, n, low, k - 1$ )
14
15 return FINDMEDIAN( $A, B, n, k + 1, high$ )
```